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THE PLETHODONTID SALAMANDER *PHAEOGNATHUS*:
COLLECTING TECHNIQUES AND HABITS

Barry D. Valentine

Known only from one specimen, the plethodontid *Phaeognathus hubrichti* Highton had consistently eluded rediscovery for almost three years. In March, 1963, a party from The Ohio State University visited the type locality in south Alabama, collected four specimens, and on the following night obtained eleven more (Figure 1A, C-D). Data from this trip are presented below, with emphasis on collecting techniques, habits, and behavior.

HISTORY.

Phaeognathus hubrichti was described as a new genus and species by Highton (1961) from the unique holotype (USNM 142486) collected in Alabama, Butler County, three mi. NW of McKenzie, just E of U. S. route 31, on 18 June 1960, by Leslie Hubricht. Since then, Valentine (in press) has redescribed the holotype, added some anatomical data, and commented briefly on the zoogeography of the region. This appears to be the entire bibliography of this remarkable salamander.

DESCRIPTION OF THE TYPE LOCALITY.

United States highway 31 runs irregularly northwestward from McKenzie, across low rolling country until it drops suddenly into the broad valley of Persimmon Creek (Figure 2A). The long straight grade down to the creek is marked by three large road cuts where the highway has bisected three ridges. The ridges strike east and west along the south side of the creek; the north face of the central ridge is the type locality of *P. hubrichti* (Figure 2B). This slope varies from about 250 to 350 feet above sea level, and Persimmon Creek itself is about 190 feet above sea level at the junction with the highway. The site is a very densely shaded, heavily wooded hillside above a small stream. The area has rich, damp leaf litter, and an abundant invertebrate fauna. Woody plants include several species of oak, at least two species of *Magnolia* (*M. macrophylla* and a larger evergreen species), sweetgum, palmetto, mountain laurel, and holly. Pines are concentrated on the ridge tops and other dryer areas, and appear peripheral to the main *Phaeognathus* habitat.

COLLECTING METHODS.

Phaeognathus does not yield to conventional collecting techniques. On the evening of 2 December 1961, I visited the type locality just before sundown, and hunted for about two hours, first in the waning light, then with a headlamp. The evening was cool enough for a jacket, but not cold, and light rain fell intermittently. Despite careful investigation of rotten logs, stones, and leaf litter, no *Phaeognathus* were seen. Other herpetologists have visited the same site, both before and after my attempt, and they too failed to find the species. This is the exact hillside where, with different collecting techniques, we subsequently found *Phaeognathus* abundant.

On 15 and 16 March 1963, four students from the Department of Zoology and Entomology at The Ohio State University (David M. Dennis, John J. Jenkinson, Steve G. Tilley, and Roger Troutman), all with herpetological interests, and myself, investigated the same hillside and after considerable fruitless search, discovered a fairly satisfactory technique for collecting the species. Fourteen of the fifteen specimens obtained were dug from the mouths of their burrows late in the evening, usually after 10:00 P. M. *Phaeognathus* is locally abundant at the type locality, and is a shy, nocturnal, highly specialized burrower. The collector walks through the woods or along the stream banks, after 8:00 P. M., and searches the ground for holes. Each hole is investigated from several angles, and when a salamander is seen, a small U. S. Army-type pick is driven into the ground behind the

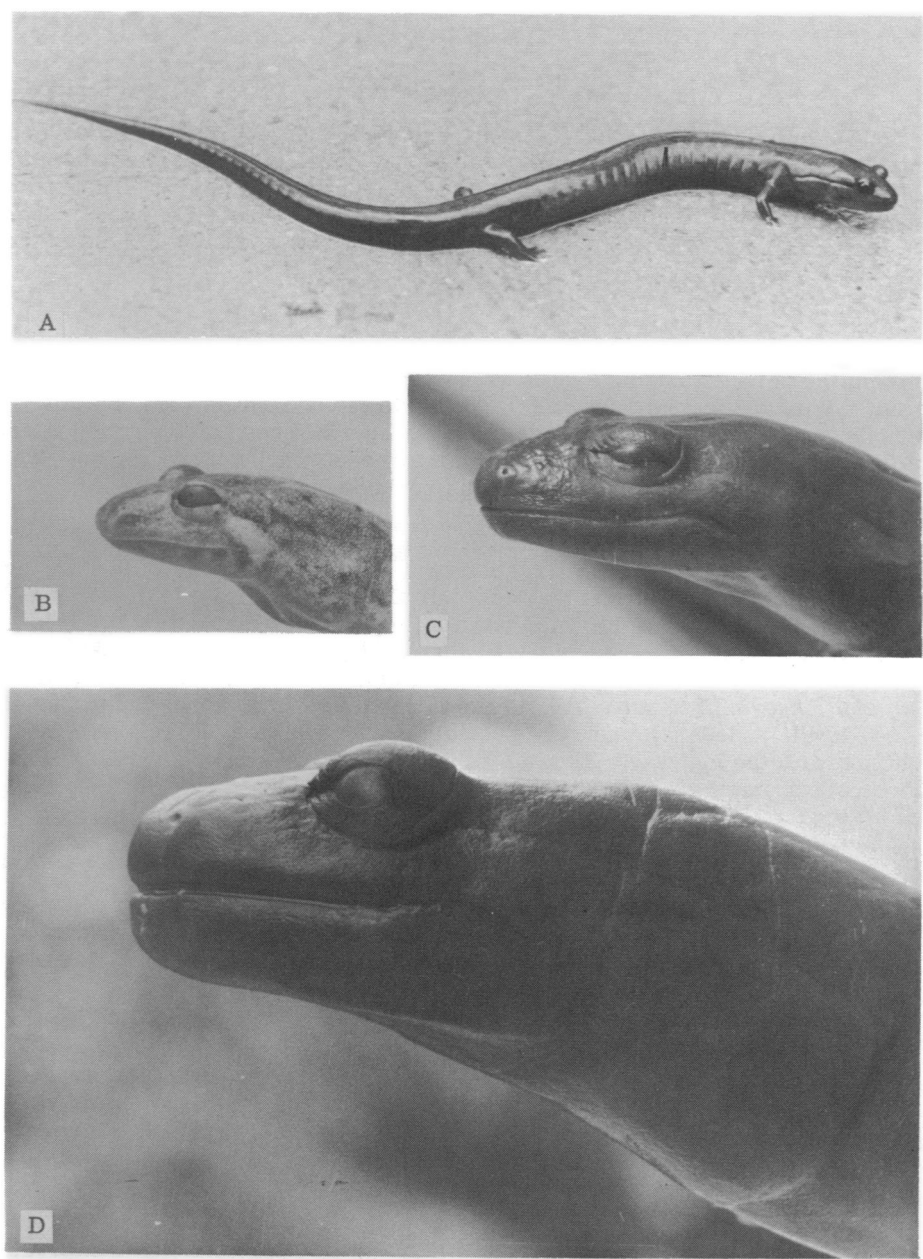


FIGURE 1. Desmognathine salamanders from Alabama, Butler County, 3 mi. NW McKenzie. A: *Phaeognathus hubrichti* Highton, living immature female to-potype, SV approximately 85 mm; B: *Desmognathus fuscus conanti* Rossman, preserved adult male, SV 46 mm; C: *Phaeognathus hubrichti*, preserved im-mature female toptype, SV 108 mm; D: Same, SV 106 mm. Figures 1B and C are to the same scale; note differences in head shape and eyelid conformation. The pustulated skin between eye and nostril in Figure 1B is due to dessication while being photographed. Photographs by the author.

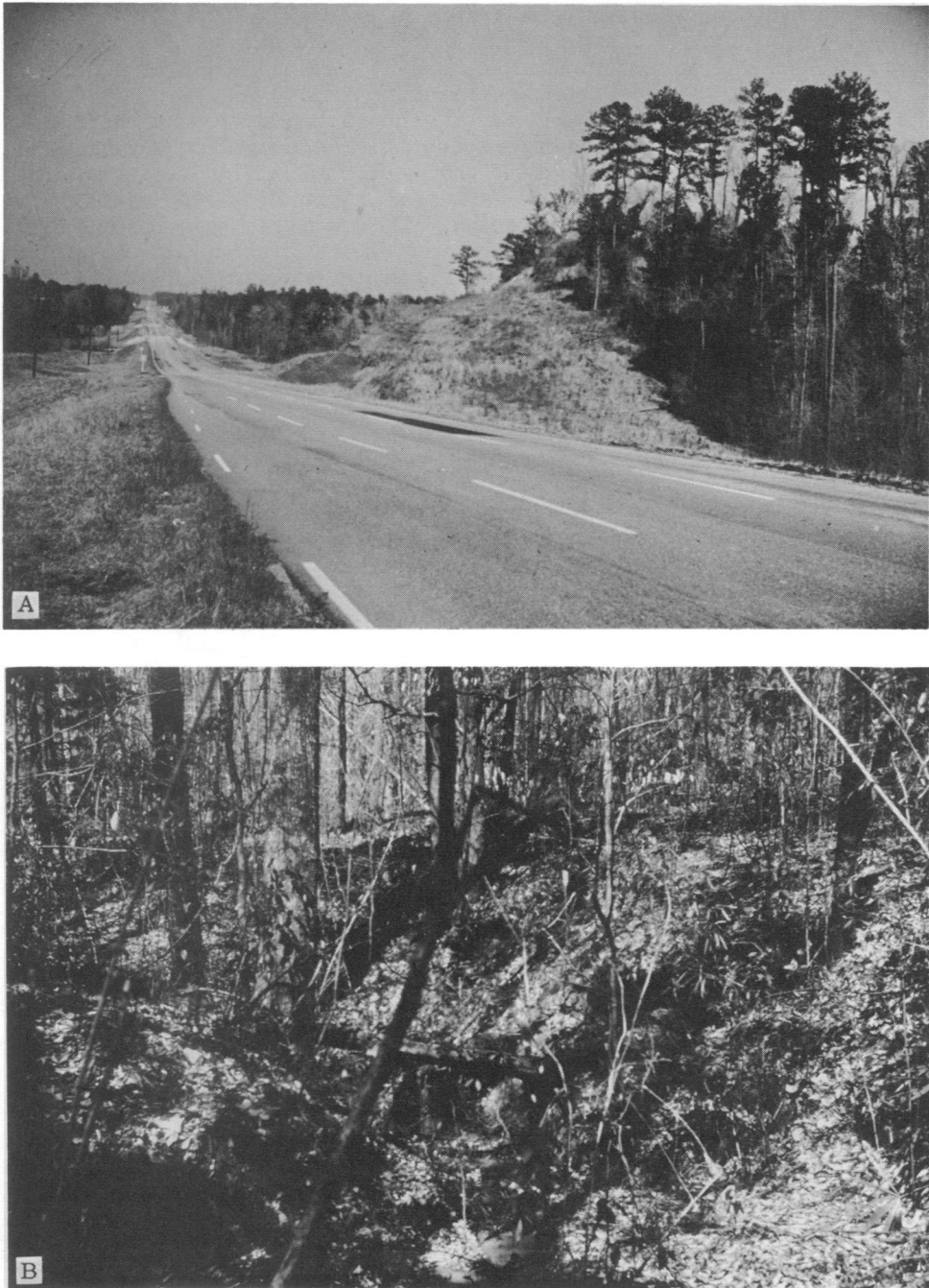


FIGURE 2. Photographs taken in Alabama, Butler County, 3 mi. NW McKenzie, 16 March 1963; the type locality of *Phaeognathus hubrichti*. A: View northward along U. S. Highway 31 into the valley of Persimmon Creek (the first of the three bridges in the background). The type locality is between the road cut visible here and the next cut which is opposite the photographer; B: View of a typical collection site at the type locality. Photographs by David M. Dennis.

animal in an attempt to prevent retreat down the burrow. We found *Phaeognathus* so shy that many escaped despite the fact that we struck with the pick and started digging as soon as a specimen was seen. Despite the almost frenzied efforts necessary for success, no specimens were mutilated during collection.

During daylight, five of us spent a total of approximately 25 man-hours of steady searching without even seeing a specimen of *Phaeognathus*. At night, after 8 o'clock, it required approximately 45 man-hours to obtain our series of 15 specimens. Thus, in 70 man-hours of intensive search, we collected 15 *Phaeognathus*, although we estimate we saw 60 specimens. The low collecting success of only one out of every four seen, was partially due to the fact that we had only one pick. The two persons who carried the pick most of the time accounted for 13 of the 15 specimens, and the other two specimens were also collected with the pick.

Mr. Leslie Hubricht writes that he found the holotype of *P. hubrichti* under the first handful of leaves he investigated. Considering the habits of *Phaeognathus*, and the hours of fruitless day-time collecting by many zoologists, the probability of such an event seems almost impossible. However, Mr. Hubricht also discovered *Plethodon richmondi hubrichti* Thurow and *P. ouachitae* Dunn and Heinze, and in the past 30 years has supplied much important data on the distribution of eastern salamanders. The many valuable specimens he has collected illustrate well the combination of originality, perseverance, knowledge of the terrain, and luck, which characterizes the superior field zoologist.

BEHAVIOR.

The inability of many herpetologists to collect *P. hubrichti* during the daytime, despite the availability of logs, stones, and leaf litter, suggests that the species stays deep in its burrow, and does not normally lie under surface cover. During the night time, the species moves to the mouth of its burrow and remains there, usually with just the head protruding although sometimes the fore legs and anterior portion of the body are visible too. Of approximately 60 specimens seen, only one was completely out of its burrow, resting on some dead leaves. All the rest backed rapidly into their burrows immediately after discovery. Some individuals retreated as soon as the light hit them, others took a few seconds before disappearing, especially if the light was turned away. No specimen remained on the surface long enough for a careful examination.

The wriggling motion used when *Phaeognathus* retreats backward into its hole is not frantic, but it is effective. On one occasion a large specimen was seen with head and shoulders protruding from a hole at the base of a tree, between two buttressed roots. There was no room to swing the pick so I lunged and caught the salamander in the neck region with my thumb and forefinger, gripping it behind the angles of the jaws. Despite my efforts, the specimen twisted and pulled free. *Phaeognathus* seems all the more unusual for after capture they are surprisingly docile. We could examine them, even turn them over and see the venter, without eliciting the violent struggles characteristic of so many other salamanders.

The single male obtained has a large semicircular scar covering five intercostal spaces on the right side. It may have been caused by the bite of another salamander. A large female, confined with a smaller individual, bit the latter's tail off near the base and apparently ate it, for no trace was found in the container. This, however, took place after 3 months in a refrigerator with little food. There was no cannibalism among ten other specimens confined together for a week; in fact, despite close quarters, none had new scars. All fifteen of our specimens had perfect tails when collected, with no sign of past injury or regeneration.

Although three large *Phaeognathus* showed no distress when submerged in Columbus tap water for 15 minutes, the species is undoubtedly as terrestrial as the larger species of *Ambystoma*. Discussion of the morphological modifications suggestive of terrestriality can be found in Valentine (in press).

HABITS.

Based on observations made in December, 1961, and March, 1963, the ravine which includes the type locality is cool and humid. On 15 March 1963, the day was very warm; however, at 5:00 P. M., just after the sun disappeared behind a ridge, the air temperature at the type locality was 74° F., and the leaf litter was 65° F. An hour later the air temperature had dropped two degrees, and the litter one. That night, the first *Phaeognathus* was seen about 7:30 P. M., another about 8:00 P. M. and then nothing until 10:45 P. M. Between 10:45 P. M. and 12:30 A. M. we saw many specimens and caught four. None were seen from 12:30 A. M. to 1:30 A. M., at which time we left the site. At the time the first specimen was collected, the temperature was cool enough for a coat, but a broken thermometer prevented further readings. The following night the sky was overcast, threatening rain, then clearing after midnight. The first specimens were seen shortly after 8:00 P. M. and by 2:00 A. M. that morning it was apparent that *Phaeognathus* was no longer at the surface. During this interval, which included a one hour break, we saw many specimens and caught eleven. During this second day, Dr. Ronald Brandon and two students from the University of Alabama, arrived at the type locality. They collected from 5:15 P. M. to 6:15 P. M. and found no specimens. Later in the evening they returned and captured three *Phaeognathus*.

We found *Phaeognathus* widespread on the north-facing slope of the ravine. Specimens were observed in their burrows from near the ridge top to within half a meter of the stream at the bottom. One specimen was collected in the next ravine to the south, but the habitat is dryer there and apparently less suitable for no others were seen. The burrow site is very variable; those easiest to find were on earth banks steep enough to be bare of litter, and those at the base of large trees. In many spots, the forest floor is almost honeycombed with holes; some of these are undoubtedly occupied by *Phaeognathus*. The origin of the holes is uncertain. At least two species of large, primitive mygalomorph spiders inhabit the area, and undoubtedly dig some of the holes. One of these, a purse web spider of the family Atypidae, is probably not too significant in this respect, for only one of their characteristic vertical tubes of silk was seen. The other spider, a trap-door species, still undetermined, is quite abundant, and it, together with smaller lycosid spiders, occupies many of the holes. Other holes of similar diameter and appearance are occupied by *Phaeognathus*. There is a strong possibility that *Phaeognathus* utilizes empty trap-door spider burrows, or perhaps vice-versa, but much more investigation is necessary. Unfortunately, digging is difficult due to the density of tree roots. Burrows of undetermined ownership occur throughout the area; those with *Phaeognathus* usually face downhill and are in areas that are damp but not seepy. *Phaeognathus* is less slimy than most plethodontids, and areas of relatively constant humidity are probably preferred over those with alternating wet and dry periods.

Several species of salamanders are associated with *Phaeognathus*. *Plethodon glutinosus* (Green) is the closest ecologically, but there are striking differences. The slimy salamander can be found under logs, even during daylight. After 8:00 P. M., both species appear, but under the conditions of our visit, *Plethodon* leaves its burrow and wanders about, some even climbing 30 cm or more on tree bases. Only one of the *Phaeognathus* seen was out of its burrow; the rest had at most the forepart of the body exposed. After about 12:30 A. M., *Phaeognathus* disappeared; however, a few *Plethodon* were still abroad at 2:00 A. M. In the bright light of a head lamp, *Phaeognathus* withdraws immediately into its burrow; most *Plethodon* remained motionless in the light, occasionally one would retreat. *Eurycea bislineata* (Green) and *E. longicauda guttolineata* (Holbrook) were common in the woods at the bottom of the ravine, near the stream. Many specimens were seen walking over the leaf-litter and crawling on steep, dirt banks. It is our impression that both species appear earlier than *Phaeognathus* - appearing at dusk and retiring at about 10:00 P. M. Several clutches of eggs and a number of larvae of *E. bislineata*

were collected in the stream. *Desmognathus fuscus conanti* Rossman (Figure 1B) is the closest relative of *Phaeognathus* at the type locality, and the most remote ecologically. *Desmognathus* is strongly aquatic, and all our specimens were found in or along the edges of the stream. Two or three were dug from burrows just above the water line but none were encountered in the woods. The second Gulf Coast species in the genus, *D. auriculatus* (Holbrook) (see Valentine, 1963) has not yet been taken at the *Phaeognathus* type locality. It is even more aquatic than *D. fuscus*, and usually does not occur in cool ravine situations.

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NOTES ON BROODS OF VIRGINIA V. VALERIAE BAIRD & GIRARD IN OHIO.

Conant (1951, *Reptiles of Ohio*, Amer. Midland Nat., 284 p.) recorded two broods of *Virginia v. valeriae* in Ohio. A female from Pike County measuring 218 mm (all measurements are total length) contained four large eggs. The other, 222 mm in length from Scioto County, had five ripening eggs in the abdominal cavity, each clearly showing an embryo. These two dissected females contained premature broods and represent the extent of our knowledge of the reproduction of this species in Ohio.

Over the past several years I have accumulated the following records of broods from four adult female *V. v. valeriae* collected in Scioto County, Ohio. All measurements were made after preservation. These specimens have been donated to Joseph T. Collins (Cincinnati) and are preserved in his collection (JTC).

I collected the first female on County Rd. 5, Shawnee State Forest, 30 August 1960. This snake gave birth to six young 13 September 1960. Only three young were preserved (JTC 419). These measured 92, 88, 9 and 85.7 mm in length.

In 1961 George McDuffie and Leonard Praeger collected two gravid females in the same forest off U.S. rt. 125, north of Boy Scout Camp Oyo. One, measuring 241.3 mm, was taken 29 July and gave birth to 9 young 26 August. The second, 215.9 mm, was captured 9 August and on 1 September gave birth to five young. Unfortunately, these two broods were mixed in the preserving pan and, as a result, the data must be given in combination. Two of the juveniles were badly decomposed and could not be measured. Measurements of the young are: Range 88.9 to 103.2 mm; Mean 96.6 mm. The above specimens (2 adult females, 14 young) are preserved together (JTC 420).

On 3 August 1962, I obtained a 266.7 mm female in Shawnee State Forest at Camp Oyo, just off U.S. rt. 125. This individual subsequently died on 16 August 1962, and contained four embryos.

In summary, the 17 specimens mentioned above, plus the 18 noted by Conant (op. cit.), bring the total known *V. v. valeriae* in Ohio to 35 (excluding embryos). In addition, the smallest recorded new-born snake (85.7 mm) is 39.3 mm shorter than Conant's smallest snake which measured 125 mm.

I should like to thank Joseph T. Collins for help in the preparation of this note and for checking my measurements.

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